

Arpit Yadav

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EDUCATION

Thapar Institute of Engineering & Technology (TIET)

Bachelor of Technology in Electrical and Computer Engineering

Patiala, India

Aug. 2023 – May 2027

TECHNICAL SKILLS

Languages: Java , JavaScript, TypeScript, Python, SQL

Frontend: React.js, Next.js, Leaflet.js, HTML5, CSS3

Backend: Node.js, Express.js, FastAPI, REST API Design, JWT, RBAC, SSE

Databases: MySQL, MongoDB, PostgreSQL, Query Optimization, Indexing

DevOps & Tools: Git, GitHub, Docker, Vercel, Render

Core Concepts: Data Structures & Algorithms, System Design, OOP, Low-Level Design (LLD)

EXPERIENCE

Freelance Full-Stack Developer

Contract

Remote

Jan. 2026 – Mar. 2026

- **BSA Design Studio** | [Live](#) - Engineered a production-ready portfolio site using React.js with reusable component architecture and image optimization; implemented responsive layouts and performance-tuned asset delivery pipelines for real-world client deployment.
- **The Seekers International** | [Live](#) - Built a Next.js + Sanity CMS portal with structured content schemas and server-side data-fetching; reduced page load latency by **35%** through query optimization and static generation strategies.

PROJECTS

FlowGrid | *React.js, Node.js, Express.js, MySQL, JWT, REST API* | [GitHub](#) | [Live](#)

- Developed full-stack service booking platform with **role-based access control** (admin, provider, customer) using **JWT authentication**, protected REST API routes, and **Razorpay payment gateway** integration supporting card, UPI, and QR-based checkout flows.
- Architected **RESTful APIs** across booking, scheduling, and payment modules with reusable **Express.js middleware** for authentication and authorization; implemented **context-based state management** and centralized API service layer in React.js for scalable frontend-backend communication.

Phatak Radar | *Node.js, Express.js, MongoDB, Leaflet.js, SSE, REST API* | [GitHub](#) | [Live](#)

- Engineered a **real-time geospatial monitoring system** tracking **145+ railway crossings** across 8 states; implemented **server-side simulation engine** with 5-second tick cycle using **Haversine distance calculations** and **MongoDB 2dsphere geospatial indexing** for proximity-based automatic gate status updates.
- Designed **RESTful APIs** for viewport-bound train queries, route polylines, and phatak state management; built **layered map architecture** with Leaflet.js rendering dynamic train markers, delay-coded overlays, and real-time event feeds via **frontend polling** on a decoupled React + Node.js stack.

Auto Insights | *React, Node.js, Express.js, FastAPI, Python, Gemini API* | [Live](#)

- Engineered a **3-tier analytics platform** (React + Node.js/Express + FastAPI microservice) with **JWT authentication & RBAC** and **sandboxed code execution engine** to run AI-generated **schema-aware SQL and Pandas code** against user-uploaded CSV/XLSX datasets.
- Integrated **Gemini API** for automated **EDA, hypothesis generation, and executive summaries**; deployed across **3 independent services** (Vercel + Render) with **PDF report** export.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Institution's Innovation Council (IIC)

Joint Secretary

TIET, Patiala, India

Aug. 2023 – Present

- Organized innovation-driven events and hackathons, collaborating with student teams, startup founders, mentors, and industry stakeholders.
- Managed a cohort of **150+ students**, fostering interdisciplinary collaboration for entrepreneurship and product development initiatives.